Towards autotopies of normed composition algebras in algebraic Quantum Field Theory

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Abstract

In a recent algebraic approach to Quantum Field Theory [FureyHughes2022-SymmBr], Nichol Furey and Mia J. Hughes demonstrate a "quaternionic triality Higgs" sector that is modeled from algebraic triality in the normed division algebras over the reals, specifically, in quaternions and octonions. Here we offer nonunital normed composition algebras with triality as additional mathematical tool, including para-quaternions and Okubo algebra. Understood as isotopes of the normed division algebras, the group of autotopies of the octonions, Spin (8), is advertised as the unifying mathematical construct. We argue for conceptual simplicity when modeling both Standard Model particle and spacetime symmetries in these algebras, which requires reconstructing conventional spacetime algebra. We give a physical motivation in terms of anthropocentric bias, a candidate mathematical mechanism in the form of a continuous Wick rotation, and refer to phenomenological support of this approach towards gravity.

Keywords: Algebraic quantum field theory, triality, Standard Model of particles, linearized gravity, reconstructing general relativity, normed division algebras, quaternions, octonions, composition algebras, nonunital algebras, Okubo algebra, para-Hurwitz algebras.

1 Context: algebraic Quantum Field Theory.

Algebraic approaches to Quantum Field Theory aim at establishing conceptual simplicity in the description of physical law, by representing empirically proven behavior of fundamental particles in terms of laws that stem from the mathematical nature of numbers themselves. In a recent paper [FureyHughes2022-SymmBr], the authors demonstrate a new representation of the Standard Model's Higgs sector, by employing quaternionic triality [Hughes2016]. They provide an explicit demonstration in the context of a single generation of Standard Model particles. Their approach permits spacetime representation through the conventional $\mathfrak{so}(3,1)$ Lie algebra, from a commuting 6-dimensional real subspace of the complex quaternions $\mathbb{C} \times \mathbb{H}$ [FureyHughes2022-OneGen].

2 Triality in nonunital composition algebras.

In the choice of a vector space V over the reals $\mathbb{R}$, *triality* f is an order 3 automorphism, i.e. a nondegenerate trilinear map from vectors $a, b, c \in V$ into the reals,

$$f : V \times V \times V \to \mathbb{R}, \tag{2.1}$$

where $f(a, b, c) = 0$ holds for any value of two of the arguments exactly when the third argument is 0. In section 3 of [FureyHughes2022-SymmBr], the authors follow the cited reference [Baez2002] and recast this as a bilinear map m,

$$m : V \times V \to V, \tag{2.2}$$

understood as a multiplication rule in an algebra $\mathbb{A}$ over vectors V. Next to requiring nondegeneracy of f, this derivation step requires existence of a multiplicative unit element, $1 \in \mathbb{A}$, which makes $\mathbb{A}$ a unital algebra, where for any $a \in \mathbb{A}$ there is

$$1 \cdot a = a \cdot 1 = a. \tag{2.3}$$

There are, however, nonunital algebras that permit triality as well [Elduque1996]. These include the 2-dimensional para-complexes $p\mathbb{C}$, the 4-dimensional para-quaternions $p\mathbb{H}$, the 8-dimensional para-octonions[1] $p\mathbb{O}$, and the 8-dimensional Okubo algebra $\mathcal{O}$. These can be understood as isotopes of normed division algebras ("Hurwitz algebras") of the same dimensionality: Given a product $\cdot$ in the complex numbers $\mathbb{C}$, quaternions $\mathbb{H}$, and octonions $\mathbb{O}$, the para-Hurwitz algebras $p\mathbb{C}$, $p\mathbb{H}$, and $p\mathbb{O}$ use a product $\circ$ that can be defined from the complex conjugates of their respective Hurwitz-counterpart:

$$a \circ b := \overline{a} \cdot \overline{b}. \tag{2.4}$$

The product $*$ of the 8-dimensional Okubo algebra $\mathcal{O}$ uses an additional order 3 automorphism ρ,

$$a * b := \overline{a\rho} \cdot \overline{b\rho^2}, \tag{2.5}$$

for example, with ρ as permutation of a select associative octonion basis element triple.

The product choices for these algebras $\mathbb{A} \in \{\mathbb{C}, \mathbb{H}, \mathbb{O}, p\mathbb{C}, p\mathbb{H}, p\mathbb{O}, \mathcal{O}\}$ are far from unique. The autotopy group of $\mathbb{O}$ is the group of isotopies of the octonions, which maps between any of the $\{\mathbb{O}, p\mathbb{O}, \mathcal{O}\}$ and their possible product choices. This group, Spin (8) [Elduque2000], is much larger than the respective subgroups of algebra automorphisms, namely, of the octonions, G_2, para-octonions, G_2, or Okubo algebra, SU(3).

Just as is the case for the normed division algebras, all nonunital algebras with triality are (nonunital[2]) composition algebras: Any such algebra $\mathbb{A}$ is equipped with a nondegenerate[3] quadratic form $\| \, \| : \mathbb{A} \times \mathbb{A} \to \mathbb{R}$, such that for any $a, b \in \mathbb{A}$ there is

$$\|ab\| = \|a\| \, \|b\| \tag{2.6}$$

(for the respective algebra product on the left-hand side, $\{\cdot, \circ, *\}$). This form is the Euclidean norm of a d-dimensional vector $a = (a_{0,} \ldots, a_{d-1})$,

$$\|a\| := \sum_{i=0}^{d-1} a_i^2. \tag{2.7}$$

Since this satisfies the triangle inequality,

$$\|a + b\| \leq \|a\| + \|b\| \,, \tag{2.8}$$

these algebras are nonunital normed composition algebras.

Appendix A very briefly traces how interest in physics led to exploring Okubo algebra for modeling the strong force during the 20th century. For contemporary mathematical interest see e.g. [Elduque2000, Elduque2022, SmithVojtech2022]. For a recent advert towards renewed interest in physics see e.g. [MCZ2023].

Note that our choice of vector space V to be over the reals $\mathbb{R}$ loses some generality. For example, special cases for vector spaces over fields with characteristic 2 or 3, or certain 2-dimensional field extensions, are omitted here.

3 From automorphisms to autotopies.

Since multiplication in the para-quaternions $p\mathbb{H}$ is modeled from mere conjugation of factors of the quaternion product, their automorphisms are the same as for quaternions, $\mathbb{H}$. This would, naively, allow understanding triality in $p\mathbb{H}$ in terms of Lie algebra, just as in the $\mathbb{H}$ case:

$$\mathrm{tri}\,(p\mathbb{H}) \cong \mathrm{tri}\,(\mathbb{H}) \sim \mathfrak{su}\,(2) \oplus \mathfrak{su}\,(2) \oplus \mathfrak{su}\,(2)\,. \tag{3.1}$$

The power of the line of research in [FureyHughes2022-SymmBr] lies in the mathematical ability to let a given algebra act on itself. Since all representations of particles and their states can be demonstrated explicitly in $\mathbb{H}$ already, there would be little to no gain if one were to suggest introducing $p\mathbb{H}$ to certain components of the model: Symmetries and their representations are fully understood in terms of Lie algebra, which can be represented in $\mathbb{H}$ (and not $p\mathbb{H}$). Introducing $p\mathbb{H}$ naively would therefore be unneeded, and violate conceptual simplicity.

One could take a slightly bolder stance, and propose to represent *all* particles and states in terms of nonunital composition algebras. With the automorphism groups of $p\mathbb{O}$ and $\mathcal{O}$ being G_2 and SU(3), respectively, it seems plausible that there is enough similarity in mathematical structure to permit this[4]. Assuming that this would indeed be possible, it might still only be of academic interest: While the algebraic approach in [FureyHughes2022-OneGen, FureyHughes2022-SymmBr] (and related works cited therein) is radical in terms of representing the Standard Model particle content algebraically,

[1] Para-octonions $p\mathbb{O}$ are sometimes also called "pseudo-octonions". They contain $p\mathbb{C}$ and $p\mathbb{H}$ as subalgebras, $p\mathbb{C} \subset p\mathbb{H} \subset p\mathbb{O}$.
[2] The term "composition algebra" is historically used with an implied a multiplicative unit, e.g. [Jacobson1958, Jacobson1981].
[3] The associated bilinear map $m\,(a, b) := \|a + b\| - \|a\| - \|b\|$ is nondegenerate, i.e., $m\,(a, b) = 0$ for any a exactly when $b = 0$, and vice versa.
[4] As is the spirit of [MCZ2023].

it is conservative in its treatment of spacetime. Just as in conventional Quantum Field Theory, spacetime symmetry remains trivially separated into its own $\mathfrak{so}(3,1)$ representation. This can be modeled faithfully in terms of Lie algebra, from a commuting 6-dimensional real subspace of $\mathbb{C} \times \mathbb{H}$. The normed division algebras can do this all, as shown, and the nonunital composition algebras with triality would again introduce unneeded complexity.

Here we advertise to consider members of the *autotopy group* of the normed division algebras for representing Standard Model particles *as well as* spacetime symmetry. Due to the large amount of historical context over the past 60 years, conflicts between diverging schools of thought that are often fought out in public without reasonable footing, open questions, and valid concerns, we restrict ourselves to presenting supporting arguments and pointers, and consciously refrain from making concrete suggestions in the direction of the Standard Model at this point. We will, however, give a concrete suggestion towards gravity.

3.1 Conceptual simplicity.

Closely following the argument towards algebraic quantum theory in [FureyHughes2022-OneGen, FureyHughes2022-SymmBr], the authors arrive at modeling physics from algebra automorphisms of the normed division algebras $\mathbb{C}$, $\mathbb{H}$, and $\mathbb{O}$ in terms of left- and right-actions of these algebras on themselves[5]. Here we have identified the nonunital normed composition algebras to be likewise allowable under their argumentation, in particular, for their proposal of quaternionic triality of the Higgs mechanism. This would add the para-Hurwitz algebras $p\mathbb{C}$, $p\mathbb{H}$, and $p\mathbb{O}$, together with the Okubo algebra $\mathcal{O}$, as additional candidates.

In order to address these additional choices, one may either rule them out by requiring a multiplicative unit element, or embrace them by understanding members of their autotopy group $\mathrm{Spin}(8)$ as representing physics. As the group of isotopies of the octonions $\mathbb{O}$, the autotopy group maps between all of these algebras[6], as well as, their respective automorphisms. In order to maintain conceptual simplicity, this approach would need to represent not only Standard Model particles, but also symmetries of spacetime.

3.2 Reconstructing spacetime.

Conventional Quantum Field Theory as the contemporary understanding of the Standard Model of physical particles imposes strong constraints on the ways particle symmetries may mix with spacetime symmetries. Known as the "Coleman-Mandula theorem", it provides a benchmark that any violation must explain itself against. In this proposal here, where spacetime may not be modeled in terms of Lie algebra anymore, we must provide reasoning for why this is motivated from physics, and propose a method how it can be achieved mathematically. With the Standard Model covering the electromagnetic, weak, and strong force, this violation of the Coleman-Mandula theorem may then either relate to new physics outside the Standard Model, or simply model the fourth remaining fundamental force of nature, gravity.

3.2.1 Physical motivation: anthropocentric bias.

We follow a well-known argument, that contemporary description of the Standard Model has been driven in part by anthropocentric bias. Observations that are more foreign to a human's day-to-day experience are modeled as variations of the more familiar Lorentzian geometry that is fundamental to electromagnetism. Made from molecules and familiar with light, human direct observation of reality is dominated by the electromagnetic force (together with a primitive experience of gravity). The inner workings of fundamental particle interactions, in contrast, are outside human direct comprehension. They may well be modeled conventionally in ways that effectively reconstruct some physical law as a projection onto Lorentzian spacetime, where such physical law would be fundamentally different in its more natural, native formulation. Conceptual simplicity then becomes a key measure when arguing for a formulation to be "native" or "natural".

This line of reasoning is of course nothing new, we just state it for clarity. We defer the reader to the wealth of proposals over the past decades, that discuss their respective violation of the Coleman-Mandula theorem. A conclusion we draw from our argument here is to keep Lorentzian spacetime foundational for electromagnetism.

[5] Using actions of algebras including $\mathbb{R} \otimes \mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$ in this context was first proposed by Geoffrey M. Dixon (e.g. [Dixon1994]), with varying models advocated over time, including [Dixon2014]. The program investigated here distinguishes itself from Dixon's by its strict focus on the given algebra, without further tensoring or matrix enlargements, to focus on modeling known properties in physics. That is, conceptual simplicity is enforced by not allowing extraneous unexplained or unobserved degrees of freedom, or unclear mixing effects from representations that use algebras much larger than what is needed.

[6] Subalgebras are mapped as long as they have the same dimensionality, of course.

3.2.2 Mathematical proposal: continuous "Wick rotation".

A commonplace mathematical tool in conventional Quantum Field Theory is Wick rotation, which changes the signature of Lorentzian spacetime to a 4-dimensional Euclidean signature, by means of transforming the time parameter t into an imaginary counterpart t' through multiplication with an imaginary unit i, i.e., $t' := it$.

Rather than complexifying time in this manner, one could just as well introduce a real phase α that transitions between Lorentzian and Euclidean spacetimes, in a way that only distinct values of α permit an understanding in terms of conventional spacetime algebra. By postulate, it would be a property of nature that is in support of the argument in this paper, as long as everything can be expressed concisely in the normed composition algebras. It would elevate Wick rotation from being a mere mathematical tool, to becoming an operation that directly derives from a postulate of the physical model.

3.2.3 Phenomenological support for quantum gravity

As a concrete example, a toy model towards such a phase is demonstrated in [Koepl2023]. There, elastic scattering of a spin-$\frac{1}{2}$ point mass against a fixed target is executed in a deliberately primitive way, using a phase $\alpha \in \mathbb{R}$ that exhibits the properties required here. This induces a rotation ϕ in complex space $\mathbb{C}_0$ to basis $b_{\mathbb{C}_0} = \{1, i_o\}$,

$$\phi := e^{i_0 \alpha}. \tag{3.2}$$

Using Pauli spinors σ_j with $j = 1, 2, 3$, represented in a different complex space $\mathbb{C}$ to basis $b_{\mathbb{C}} = \{1, i\}$,

$$\sigma_1 := \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 := \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_3 := \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \tag{3.3}$$

and unit matrix

$$I_2 \equiv \sigma_0 \quad := \quad \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \tag{3.4}$$

conventional Dirac spinors are generalized to some $\gamma_\mu \in \mathbb{C}_0 \oplus \mathbb{C}$ in matrix form:

$$\gamma_0 = \begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix} = \begin{pmatrix} I_2 & 0 \\ 0 & -I_2 \end{pmatrix}, \qquad \gamma_j = \begin{pmatrix} 0 & \phi^2 \sigma_j \\ \sigma_j & 0 \end{pmatrix}. \tag{3.5}$$

Together with the identity 4×4 matrix I_4, these generalized Dirac matrices satisfy:

$$\frac{1}{2}\left(\gamma_\mu \gamma_\nu + \gamma_\nu \gamma_\mu\right) = I_4 * \begin{cases} 0 & \text{for } \mu \neq \nu, \\ 1 & \text{for } \mu = \nu = 0, \\ \phi^2 & \text{otherwise } \left(\mu = \nu \in \{1, 2, 3\}\right). \end{cases} \tag{3.6}$$

Only in the special cases $\phi^2 = 1$ and $\phi^2 = -1$ can the γ_μ still be viewed as spinors, on which a conventional spacetime algebra with Euclidean or Lorentzian signature may act.

This ansatz yields effects that one may expect from a model of quantum gravity: Similar to scattering a test body in Schwarzschild metric on the central mass, elastic backscattering of a spin-$\frac{1}{2}$ quantum particle on a fixed target is enhanced here, and the scattering cross section is nonzero in the limit for high energies. Conversely, in known processes in nature (neutrinos moving through matter, scattering in intergalactic gas), the estimated momentum transfer is shown small enough to not yet have been noticed.

3.2.4 Anthropocentric reconstruction of General Relativity

Leaving the quantum world, projection of an $1/r$-type potential from an assumed 4-dimensional Euclidean spacetime into Lorentzian observer spacetime is a way to reconstruct General Relativity in the large-body, non-quantum limit [Koepl2007][7]. Observer dynamics is proposed to be modeled from a Lorentz-invariant equation of motion, where for

[7] In that paper, it is sufficient to ignore the auxiliary complex-octonion setting, and to only pick the self-contained argument in section 4 proposition 4 (the "*NatAliE* equations"). Also note that equation (56) gives a factor $\sqrt{1 - |\vec{v}|^2}$ in the denominator, which is explained away later on with words. This factor is erroneous, and should not have been introduced here from starters: Only the body's invariant mass should have been projected out from Euclidean geometry, and not its effective mass after Lorentz transformation.

$\mu, \nu = 0, 1, 2, 3$ there are charges $M_{\mu\nu}$ that effect potentials $h_{\mu\nu}$ from second order spacetime differentials $\partial_\mu^2 \equiv \partial^2/\partial x_\mu^2$, $t \equiv x_0$, $\vec{x} \equiv (x_1, x_2, x_3)$:

$$\Box h_{\mu\nu} \equiv \left(\partial_0^2 - \sum_{i=1}^{3} \partial_i^2\right) h_{\mu\nu} = -8\pi M_{\mu\nu}. \tag{3.7}$$

This can be solved in closed form, known as "retarded potentials":

$$h_{\mu\nu}(t, \vec{x}) \quad = \quad -2 \int d^3 r \frac{M_{\mu\nu}(t - |\vec{r} - \vec{x}|, \vec{x})}{|\vec{r} - \vec{x}|} \tag{3.8}$$

The Lorentzian observer takes an anthropocentric view and assumes all physical bodies in moving frames of reference to be subject to apparent length contraction and increased masses. Gravity, in contrast, is proposed to be governed by a 4-dimensional Euclidean native geometry. In such a geometry, a physical body that is moving at a relative speed $|\vec{v}|$ should appear with lengths $|\vec{x}'|$ expanded,

$$|\vec{x}| = |\vec{x}'| \sqrt{1 + |\vec{v}|^2}, \tag{3.9}$$

and invariant mass m' reduced,

$$m = \frac{m'}{\sqrt{1 + |\vec{v}|^2}}. \tag{3.10}$$

These adjustments are projected into the (Lorentzian) retarded potential of a single point mass, to satisfy the anthropocentric view of the Lorentzian observer. Since the resulting tensor formulation is linear, it allows summation across any number $l = 1, 2, 3, \ldots$ of point masses m_l by means of the combined energy-momentum tensor $M_{\mu\nu}$. Together with Minkowski tensor $\eta_{\mu\nu}$ and total mass $\rho := \sum_l m_l$, the general equation of motion becomes

$$\Box h_{\mu\nu} = -16\pi\gamma \left(M_{\mu\nu} - \frac{1}{2}\rho\eta_{\mu\nu}\right), \tag{3.11}$$

and can be identified as the canonical linear approximation of General Relativity.

A well-known "bootstrap" argument then warrants that covariant General Relativity can be recovered from such linearization, by progressively undoing the approximations that led to it: Starting at the fields generated from linearized gravity, it reconstructs generally covariant field self-interaction iteratively, by treating the fields at each iteration as new sources of gravity. As such, it is a faithful reconstruction of General Relativity from a flat 4-dimensional real spacetime.

In this semiclassical (large body, non-quantum) model, nature's postulated native geometry would contain Lorentzian and Euclidean spacetimes in distinct cases. The rich structure in terms of the autotopy group of the normed composition algebras, as proposed in this paper, would then correspond to generalization to the quantum world by means of a continuous phase for gravity, and an algebraic Quantum Field Theory for the forces of the Standard Model of particles.

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A Okubo algebra in 20th-century physics

There is a direct historical trace from particle physics to nonunital composition algebras. One of Susumu Okubo's celebrated achievements in the 1960s is known today as the "Gell-Mann-Okubo mass formula" [WP:GellMannOkuboMass]. It is a key discovery towards contemporary understanding of the strong force in physics, in terms of quarks and a field theory with $\mathrm{SU}(3)$ gauge symmetry.

Okubo went on to propose an algebra that has $\mathrm{SU}(3)$ as is automorphisms, in terms of a deformed matrix product: Given two trace-0 complex 3×3 matrices a, b and scalars $\lambda, \mu \in \mathbb{C}$ satisfying $\lambda + \mu = 3\lambda\mu = 1$, Okubo introduced the product

$$a * b := \lambda ab + \mu ba - \frac{1}{3}\mathrm{Tr}(ab). \tag{A.1}$$

With the group of structure-preserving, invertible mappings between products of this sort being the same as observed symmetries in physics, $\mathrm{SU}(3)$, this algebra was explored during the remainder of the 20th century for modeling the strong force. For pointers see e.g. [Okubo1995]. The modern term "Okubo algebra" was established later, well after the relevant publications.

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